



Review Article

Advancements and trends in mangrove species mapping based on remote sensing: A comprehensive review and knowledge visualization

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ABSTRACT

Mapping and monitoring of mangrove species based on remote sensing technology play a crucial role in biodiversity conservation and management. This paper employs CiteSpace to visualize the literature and presents a comprehensive review of the researches conducted in this domain, focusing primarily on bibliometric characteristics, diverse sensors, and classification algorithms. Since the publication of the first remote sensing-based study on mangrove species classification in 2004, the number of publications in this field has exhibited a general upward trend up to 2023. China, the United States, and India lead in publishing research on mangrove species mapping, with researchers in the United States being particularly active in international collaborations. Mapping of mangrove species is predominantly concentrated on single time points and across 53 small regions, with the majority of research sites located in India and China. Existing studies have utilized various remote sensing image for mangrove species classification, including airborne hyperspectral, spaceborne visible, infrared, multispectral, hyperspectral, synthetic aperture radar (SAR), and drone-borne visible, infrared, multispectral, hyperspectral, light detection and ranging (LiDAR) data. Classification algorithm development has evolved four stages, from pixel-based methods to object-oriented approaches, progressing to approaches incorporating machine learning algorithms, and currently advancing towards ensemble learning and deep learning. Research in this field still faces several challenges in data fusion, classification algorithm enhancement, increased number of classification species, and large-scale long-term mapping. The study's findings would provide valuable guidance to researchers and practitioners in advancing and enhancing the management and conservation of mangroves.

1. Introduction

Mangroves are salt-tolerant woody plants that thrive in the mudflats of intertidal regions and estuarine areas along tropical and subtropical coastlines (Brander et al., 2012). They serve as protective barriers, shielding the coastline from tidal waves and tsunamis,

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which is incredibly relevant in both ecological and economic contexts (Chang and Mori, 2021). Additionally, they provide breeding and nursery habitats for marine species while also helping to sequester carbon from the atmosphere, acting as crucial long-term carbon sinks (Donato et al., 2011; Macreadie et al., 2021). However, due to both natural and anthropogenic threats, mangroves are found to be rapidly vanishing around the world (Polidoro et al., 2010; Bryan-Brown et al., 2020). In the past 50 years, approximately 35 % of the world's mangroves have disappeared and predictions indicated that 100 % of mangrove functionality would disappear if the current rate of loss persists over the next century (Valiela et al., 2001; Friess et al., 2019).

Globally, approximately 96 % of mangroves are located in tropical regions, with Asia hosting the largest amount of mangrove cover and highest species diversity. There are about 80 mangrove species worldwide, covering about 1.45×10^5 km² of coastal zone (Tran et al., 2022). Species composition and diversity are critical parameters for health assessment, as the decline in species diversity frequently results in the deterioration of mangrove structure and function (Kacic and Kuenzer, 2022). Thus, mapping and monitoring mangrove species composition and their spatio-temporal distribution are essential for informing conservation management decisions concerning potential threats and for modeling changes in relation to driving factors (Jia et al., 2018). This, in turn, contributes to the preservation of mangrove quality and biodiversity.

Traditional mangrove mapping approaches, such as field observation and surveys, are time-consuming, laborious, and expensive. In contrast, remote sensing techniques for species-level mapping and monitoring of mangroves offer a more time- and cost-effective alternative (Palmer et al., 2002; Ismail et al., 2018). Numerous studies have recognized the effectiveness of remote sensing for mapping and monitoring mangrove species, and several review articles have provided a valuable foundation for researchers in this domain. For example, Wang et al. (2019) considered the hotspot of mangrove remote sensing research during 2000–2009 turned to species classification, with a significant increase in publications in this research field after 2011. A review study based on relevant research from 2010 to 2019 categorized mangrove species classification methods into traditional and machine learning approaches, revealing that while traditional methods, utilizing high spatial resolution imagery and object-oriented image analysis, have made significant progress in distinguishing mangrove species, their overall accuracy (OA) is not as high as that of machine learning algorithms employing diverse datasets (Pham et al., 2019b). A recent study provides a detailed overview of each algorithm used for mangrove species classification in previous publications, and put forward the advantages and limitations of the algorithm in application (Maurya et al., 2021). Although these studies describe and analyze the basic progress in remote sensing-based mangrove species monitoring and mapping research, previous literature researches in this field have been mostly descriptive and lack systematic and comprehensive analysis, particularly in objective quantitative literature research based on visual analysis.

Given the significance of mangrove species mapping and the limitations identified in previous literature analysis, the main objectives of this study are to (1) analyze the bibliometric characteristics of existing studies on mangrove species mapping and monitoring based on remote sensing, (2) summarize the various sensors and image types used for mangrove species remote sensing classification, (3) explore the characteristics and emphasis of classification techniques employed to map and monitor mangroves species at different stages, (4) identify the current limitations and challenges for studies in this field. To fulfill these objectives, literature data was gathered from the Web of Science (WoS) and China National Knowledge Infrastructure (CNKI) databases and analyzed using visualization software CiteSpace.

2. Materials and methods

2.1. Data source and literature search strategy

Articles related to mangrove mapping documented in WoS and CNKI were collected up to the year 2023. The retrieval strategy involved searching for the topics “mangrove species” and “remote sensing”. A total of 484 articles were retrieved from WoS, and 79 articles from CNKI. To ensure research accuracy, the full texts of all identified literature were reviewed. After individually screening the articles, 112 valid articles from the WoS database and 58 from CNKI were selected, resulting in a total of 170 articles. The first remote sensing-based paper on mangrove species classification was published in 2004, thus the time span for the collected literature ranges from 2004 to 2023. Full records, including authors, titles, published journals, and abstracts of the searched articles, were downloaded in the Plain Text format.

2.2. Analysis tools

CiteSpace (version 6.1.R4) stands out as a potent tool for quantifying and visualizing publications, requiring a Java environment to operate (Chen, 2006; Yu et al., 2023). Utilizing precise literature combing and dynamic visualization techniques, it can autonomously generate maps that facilitate the co-occurrence analysis of collaborative relationships among published works, such as publishing countries, authors, and keywords (Qin and Li, 2023). This enables the exploration of collaboration proximity between various nodes, thereby aiding in the investigation of research hotspots and development trends within a designated time period and scientific field. Additionally, Excel and ArcGIS were further employed for data processing and visualization in this study.

2.3. Knowledge graph analysis and descriptive statistics

The literature data, spanning from January 2004 to December 2023, were imported into CiteSpace for format conversion, deduplication, and analysis. A knowledge graph of mangrove species mapping research based on remote sensing was constructed and analyzed, focusing on countries cooperation, keywords cluster, and other relevant entities as nodes. The analysis used one-year time

slices, with a threshold of the top 25 items per slice. Using “keyword” as the note analysis type, keyword cluster analysis was performed employing similarity measures and specific clustering algorithms, i.e. the latent semantic analysis (LSA) and log-likelihood ratio (LLR). The term with the highest score in each cluster was then selected as the cluster label. Additionally, taking “country” as the note analysis type, cooperation relationships between research institutions were explored through the cooperation analysis function.

Furthermore, based on the collected full records, the annual publication volume and the journals in which the articles were published were analyzed. The research locations from the existing literature were compiled and mapped on a global scale using ArcGIS.

3. Results and analysis

3.1. Bibliometric characteristics

3.1.1. Trends in mangrove species mapping publications

Publications on mangrove species mapping and monitoring exhibited a fluctuating yet overall upward trend (Fig. 1), reflecting the increasing maturity of remote sensing-based mangrove species classification technology. It is noteworthy that the number of annual publications in CNKI has declined since 2020, likely due to the COVID-19 pandemic, which caused delays or cancellations of most field trips and impacted field data collection, along with certain effects on academic workloads (Raynaud et al., 2021). About 25.89 % of the publications were found in the journal of Remote Sensing, with others appearing primarily in the International Journal of Remote Sensing (8.04 %), Proceedings of SPIE (4.46 %), ISPRS Journal of Photogrammetry and Remote Sensing (3.57 %), Remote Sensing of Environment (2.68 %) (Table A1). These studies are predominantly associated with key subjects such as remote sensing, imaging science photographic technology, environmental sciences, geosciences multidisciplinary, and biodiversity conservation.

3.1.2. Keyword clustering and research directions

Keywords are central to articulating the core themes and research topics within academic articles, and keyword co-occurrence analysis (Table 1) is instrumental in identifying research hotspots and emerging trends. A keyword with a centrality greater than 0.1 is recognized as having significant influence. High-frequency and high-centrality keywords such as “classification”, “hyperspectral data”, “remote sensing”, and “machine learning” highlight the major research areas in this field. Furthermore, a cluster analysis of keywords using the LSA algorithm revealed 12 distinct research topics (Fig. 2). Remote sensing and machine learning were core topics of the network which exhibit a strong interrelationship. By analyzing literature in conjunction with high-frequency keywords and the 12 identified clusters, research directions can be broadly categorized into two primary areas. On one hand, a set of studies are dedicated to improving the classification of mangrove species by algorithmic modifications or novel data source utilization. On the other hand, certain studies have delved into biomass, carbon stocks or other biological parameters in relation to mangrove species classification.

3.1.3. Global collaboration relationships

Mangrove species mapping and monitoring has become a research topic of global collaboration and participation. Based on literature data obtained from the WoS database and the country cooperation analysis function of CiteSpace, the number of articles published in 32 different countries was analyzed. The number of publications indexed in the WoS can, to some extent, reflect a country’s research strength in a particular field. The network of cooperation among countries (Fig. 3) indicated that China, the United States, India, Australia, Canada, Japan, and Indonesia have a higher volume of publications in this field compared to other countries.

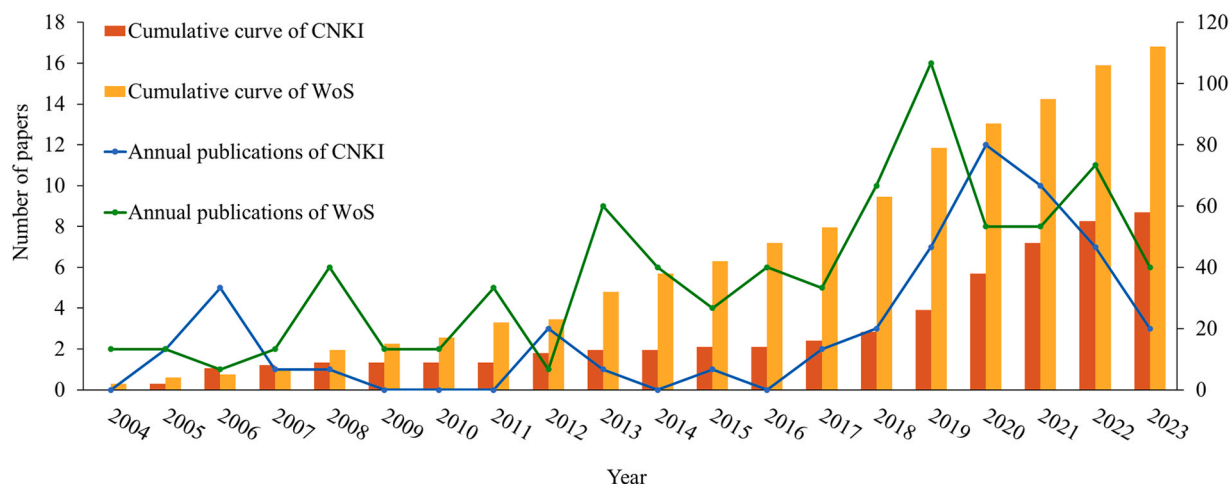


Fig. 1. Trends in publication of remote sensing-based mangrove species mapping in WoS and CNKI databases.

Table 1
Top 15 keywords with high frequency and centrality in remote sensing-based mangrove species mapping research.

Rank	High frequency			High centrality		
	Keywords	Frequency	Year	Keywords	Centrality	Year
1	Forest	45	2004	Classification	0.26	2004
2	Classification	37	2004	Forest	0.22	2004
3	Hyperspectral data	30	2013	Remote sensing	0.20	2004
4	Ecosystem	28	2007	Hyperspectral data	0.18	2013
5	Remote sensing	25	2004	Machine learning	0.16	2010
6	Machine learning	20	2010	Vegetation index	0.13	2017
7	Imagery	19	2006	Mangrove species	0.12	2004
8	Vegetation	19	2007	Feature selection	0.12	2005
9	Discrimination	13	2011	Discrimination	0.08	2011
10	Mangrove species	12	2004	Random forest	0.08	2017
11	Ikono	12	2004	Ikono	0.08	2004
12	Random forest	11	2017	Canopy	0.07	2005
13	Bioma	9	2014	Integration	0.07	2013
14	Aboveground bioma	8	2019	Bioma	0.06	2014
15	Canopy	7	2005	Future	0.05	2008

Note: Frequency indicates the number of publications in which a keyword appears; Year denotes the first appearance of the keyword; Centrality reflects the degree of the keyword's contribution to the knowledge network in mangrove species mapping.

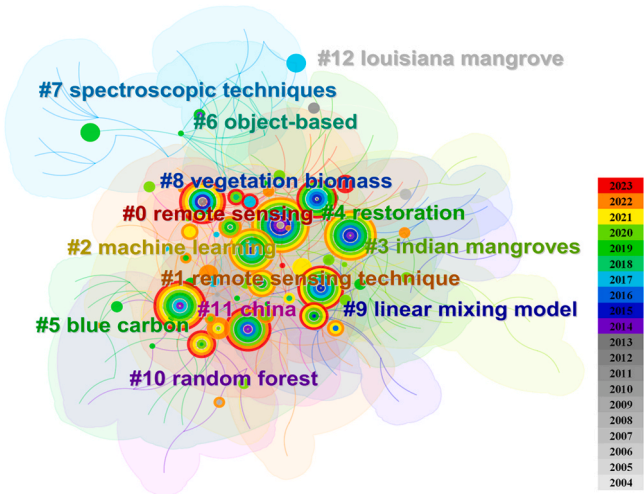


Fig. 2. Keyword clustering diagram of mangrove species mapping based on remote sensing.

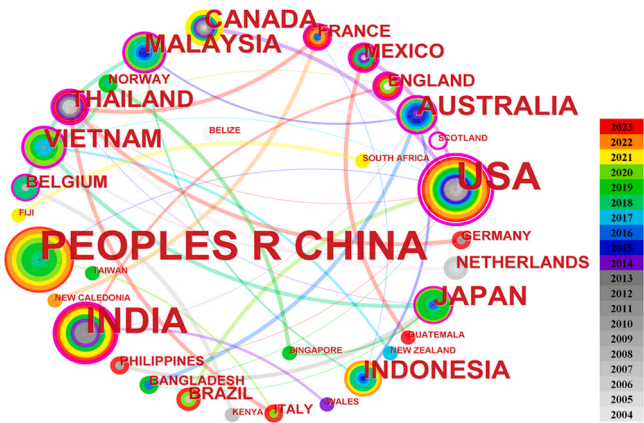


Fig. 3. Network of countries for co-authored articles (Nodes, node size, connecting lines, and line width represent the countries of author affiliations, number of published papers, cooperative relationship, and cooperative strength, respectively.).

Notably, the researchers from the United States have the most extensive international collaborations, partnering with a diverse range of countries including Mexico, Canada, China, Thailand, and Brazil. In comparison, despite the substantial volume of mangrove species mapping research in China and India, there remains significant potential for improvement in international communication and collaboration. Nodes with a betweenness centrality value above 0.1 are considered as pivot nodes, and their outer edges are shown in purple. In the visualization graph (Fig. 3), pivot nodes include the United States, Mexico, India, France, Malaysia, Thailand, Vietnam, England, Japan, Australia, and Belgium, indicating that the literature in these countries is of high quality and has greater influence.

3.2. Analysis of study implementation sites

Scholars have conducted mangrove species mapping and monitoring across various locations. Fig. 4A illustrates the previous studies were mainly carried out on mangrove sites in China, India, Indonesia, Vietnam, Malaysia, Costa Rica, and Mexico. The global distribution of mangrove research sites is mainly concentrated in two regions i.e. Atlantic-East Pacific region (Fig. 4B) as well as the Indo-West Pacific region (Fig. 4C) (Numbere, 2018; Ricklefs et al., 2006). Through extensive research on mangrove species diversity, several key research hotspot areas have been identified. Prominent among these are the Sundarbans in Indian (studied 9 times), Mai Po in Hong Kong, China (8 times), Beibu Gulf in Guangxi Province, China (5 times), and Punta Galeta in Panama (4 times). Grouped by continent, 67.92 % of the study sites are concentrated in Asia, followed by North America (13.21 %), Oceania (7.55 %), Africa (5.66 %), and South America (5.66 %). Fig. 4D shows the number of research areas with different research frequencies in each country. Sites studied 1–3 times are the most common, comprising 92.45 % of the total research areas, with the majority located in China (9 sites), Indonesia (6 sites), India (6 sites), and Vietnam (5 sites). In contrast, sites studied 4–6 times and 7–9 times are less frequent. This distribution exhibits a notable correlation with the network of international collaboration among countries (Fig. 3), reflecting similarities between research activity and geographic focus.

3.3. Data sources for mangrove species mapping

The various sensors used in mangrove species mapping are compiled in Table 2, encompassing medium (Landsat, Hyperion), high (SPOT, RapidEye, GeoEye, WorldView), and ultra-high (centimeter level) spatial resolution data obtained from satellite (Spaceborne), aircraft (Airborne), and UAV (drone) remote sensing platforms. Due to the comparability of these studies being hindered by variations in study locations, species classifications, imagery sources, and classification algorithms, a definitive consensus is lacking on the optimal data source or method for mapping mangrove species. Nonetheless, some discernible trends could still be observed. Thus, the performance of mapping mangrove species was evaluated in light of the aforementioned factors, relying on reported OA in the relevant

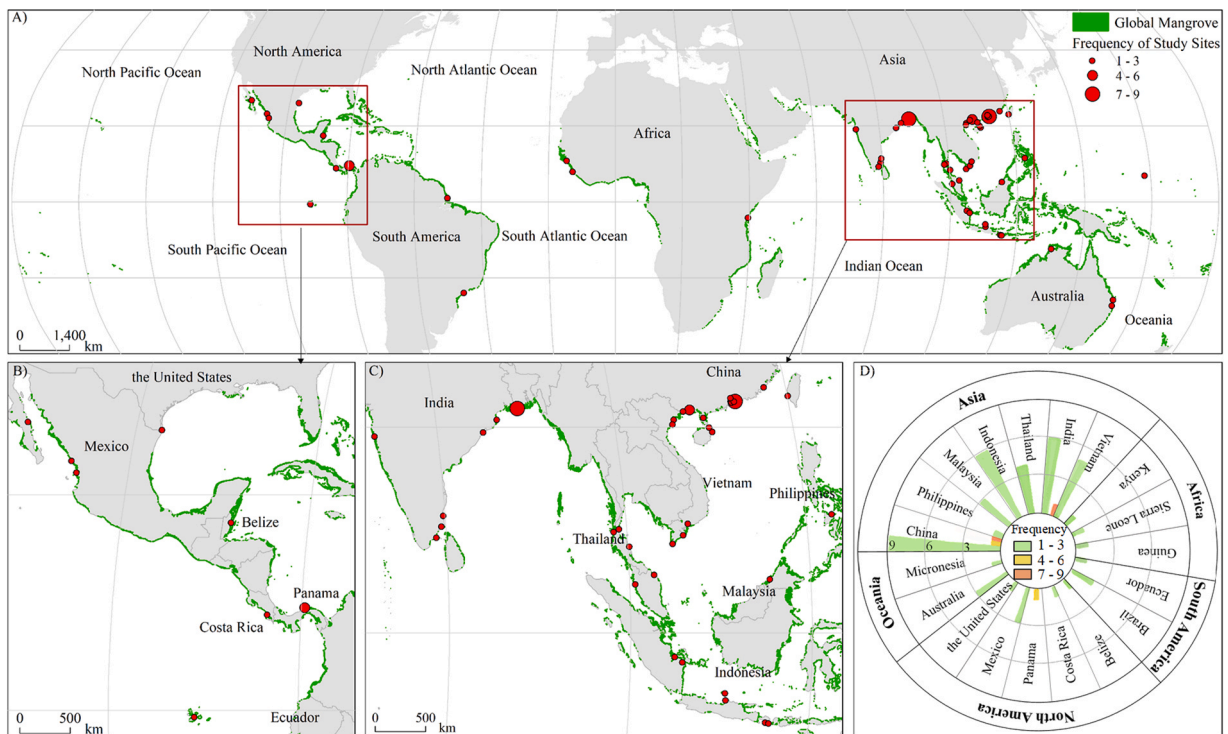


Fig. 4. Spatial distribution of remote sensing-based mangrove species mapping research locations (Global mangrove distribution data sourced from Global Mangrove Watch (Bunting et al., 2022)).

Table 2

Type of remote sensing image and sensors used in mangrove species mapping.

Remote sensing imagery	Type of image used	Sensor	Spatial resolution (m, pts/m ²)	Access	Temporal resolution	Data processing complexity	First time used
Visible spectral and infrared	Spaceborne	Pleiades–1A	0.500	Commercial	1 day	Low	2018
		Pleiades–1B	0.500	Commercial	1 day	Low	2018
	drone	SJ4000 RGB and IR cameras	0.050	Paid	Real-time	Low	2018
		Sony RX1RM2	0.020	Paid	Real-time	Low	2019
		Zenmuse X4S camera	0.050	Paid	Real-time	Low	2022
		Original DJI Phantom 4 camera	0.053	Paid	Real-time	Low	2023
Multispectral	Spaceborne	IKONOS	4.000	Commercial	3–5 days	Moderately low	2004
		QuickBird	2.400	Commercial	2.5–3.5 days	Moderately low	2004
		Landsat–7	28.500	Free	16 days	Low	2008
		IKONOS–2	4.000	Commercial	3–5 days	Moderately low	2010
		WorldView–2	1.600	Commercial	< 1 day	Moderately low	2011
		GeoEye	1.650	Commercial	< 1 day	Moderately low	2013
		SPOT–5	2.500	Commercial	2–3 days	Low	2014
		RapidEye	4.000	Commercial	5–6 days	Moderately low	2016
		WorldView–3	1.240	Commercial	< 1 day	Moderately low	2016
		SPOT–4	20.000	Commercial	2–3 days	Low	2017
		Sentinel–2	10.000	Free	10 days	Low	2017
		Landsat–8	30.000	Free	16 days	Low	2018
		Gaofen–1	8.000	Paid	4 days	Low	2018
		IRS Resourcesat–2 LISS-IV	5.800	Commercial	24 days	Moderately low	2019
		SPOT–7	6.000	Commercial	1–4 days	Moderately low	2020
		SPOT–6	1.500	Commercial	1–4 days	Moderately low	2020
	drone	Gaofen–2	1.000	Paid	1 day	Low	2020
		Gaofen–5	30.000	Paid	30 days	Low	2020
		Landsat–9	30.000	Free	16 days	Low	2023
		UHD 185	0.020	Paid	Real-time	Low	2018
		FC6360	0.058	Paid	Real-time	Low	2022
		RedEdge MX	0.066	Paid	Real-time	Moderately low	2022
		Survey3 W camera	0.055	Paid	Real-time	Low	2023
		AISA+	4.000	Paid	4 days	High	2009
Hyperspectral	Airborne	CASI	2.000	Paid	2 days	High	2011
		AVIRIS-NG	4.000	Paid	4 days	High	2019
		SASI	3.500	Paid	3.5 days	High	2021
		EO–1 Hyperion	30.000	Free	16 days	High	2013
	Spaceborne	Zhuhai–1	10.000	Free	3–5 days	Moderately high	2022
		DESI	30.000	Free	3–5 days	High	2023
		PRISMA	30.000	Free	7 days	High	2023
		Headwall Nano-hyperspec push-broom	0.100	Paid	Real-time	High	2021
	drone	Rikola camera	0.065	Paid	Real-time	High	2021
		Spectrum Patrol HSG–1	0.100	Paid	Real-time	High	2023
		ALOS PALSAR	20.000	Paid	14–46 days	High	2013
		ENVISAT ASAR	30.000	Free	30 days	High	2014
Synthetic aperture radar (SAR)	Spaceborne	ALOS AVNIR–2	10.000	Paid	46 days	High	2015
		TanDEM-X	12.000	Paid	11 days	High	2019
		RadarSat–2	3.000	Paid	24 days	High	2020
		Sentinel–1A	10.000	Free	12 days	High	2020
		Gaofen–3	1.000	Paid	8 days	High	2020
	drone	GL–70A survey-grade LiDAR system	20.000	Paid	Real-time	High	2019
		Velodyne VLP–16 puck sensor	94.000	Paid	Real-time	High	2019
		Riegl VUX–1UAV laser scanner system	16.590	Paid	Real-time	High	2021
		AS–900HL multi-platform laser scanning system	100.000	Paid	Real-time	High	2023

studies (Shen et al., 2022).

The first paper about mangrove species mapping with high accuracy were published following the launch of high-resolution satellite sensors (IKONOS), Wang et al. (2004a) successfully mapped three mangrove species in Caribbean coast of Panama, achieving an accuracy ranging from 71.3 % to 97.9 %. Since then, a vast amount of mangrove species mapping research has been carried out, with most of it based on multispectral or hyperspectral (Conchedda et al., 2008; Jiang et al., 2021). Multispectral data mainly comes from satellite and UAV platforms, while the acquisition of hyperspectral images covers three remote sensing methods: satellites, aircraft and UAV. Previous studies show that medium spatial resolution satellite remote sensing imagery such as Landsat,

with a resolution about 30 m, tend to get a lower accuracy, reaching on average OA about 65 % (Wang et al., 2018a; Valderrama-Landeros et al., 2018). When high spatial resolution remote sensing data sources such as Gaofen, GeoEye and WorldView, are utilized to map mangrove species, they often achieve better results than medium-resolution ones. The number of bands and resolution of the sensor restrict mangroves species discrimination accuracy, especially when species with similar characteristics are mixed. Within a single forest stand, the diversity of mangrove species can reach up to 58 different types (Md Isa and Suratman, 2021). While research has showcased the capability to distinguish among 16 mangrove species by utilizing six hyperspectral channels, most remote sensing studies published so far have been limited to discriminating only up to seven species, and the most commonly found genera are *Rhizophora*, *Avicennia*, *Bruguiera* and *Rhizophoropsis* (Vaiphasa et al., 2007; Mandal and Bar, 2018; Cao et al., 2021). It has been noted that as the spatial resolution increases and approaches the size of the plant canopy, the accuracy of species identification tends to improve. Consequently, an increasing number of scholars have attempted to enhance the accuracy of species mapping by incorporating high spatial resolution remote sensing data. Additionally, in the current era of digital image processing, the fusion and integration of data from multiple sensors and resolutions have garnered increasingly more attention. Fusing images derived from various remotely sensed datasets (e.g., optical, SAR) has provided [supplementary information](#), which aids in reducing uncertainties and errors in mapping mangrove species, thereby improving classification accuracy.

Most studies published before 2020 scarcely utilized UAV imagery. Nonetheless, they consistently underscored the significance of exceptionally high spatial and spectral resolution. Later, UAV remote sensing platforms have rapidly sprung up due to the centimeter-level resolution and flexible acquisition cycles (Zimudzi et al., 2021). It can not only serve as the main data source but also assist in accuracy verification for species mapping (Jones et al., 2010; Hill et al., 2017; Kattenborn et al., 2019). A mangrove species mapping research in Qi'ao Island used images obtained from UAV equipped with a UHD 185 hyperspectral sensor and achieved an OA of 88.66 % (Cao et al., 2018). Another study compared the performance of UAV and WorldView-2 (WV-2) imagery in mapping mangrove species and the results indicated that UAV (OA=94 %) was better than that of WV-2 (OA=73 %) (Jiang et al., 2021). The OA of species identification based on UAV is higher than that of some satellite data sources, which means UAV has great potential for mangrove species mapping. At the same time, UAV can not only carry independent optical sensors but can also be equipped with LiDAR systems to capture point cloud data. Studies have demonstrated that the integration of optical and LiDAR data significantly enhances species mapping accuracy, as the three-dimensional structural information provided by point cloud data has a crucial impact on species identification (Li et al., 2021). For example, in a study about mapping mangrove species in two distinct mangrove forests in China found that integrating UAV-LiDAR and Sentinel-2 data resulted in the highest OA (85.60 % and 91.61 %), while using only Sentinel-2 (80.00 % and 80.42 %, respectively) or UAV-LiDAR data (77.60 % and 75.52 %, respectively) yielded lower accuracies (Wang et al., 2022b). The results also consistent with previous studies, suggesting that image fusion brings new, better quality data compared to originals, ultimately leading to higher classification accuracy (Fu et al., 2022).

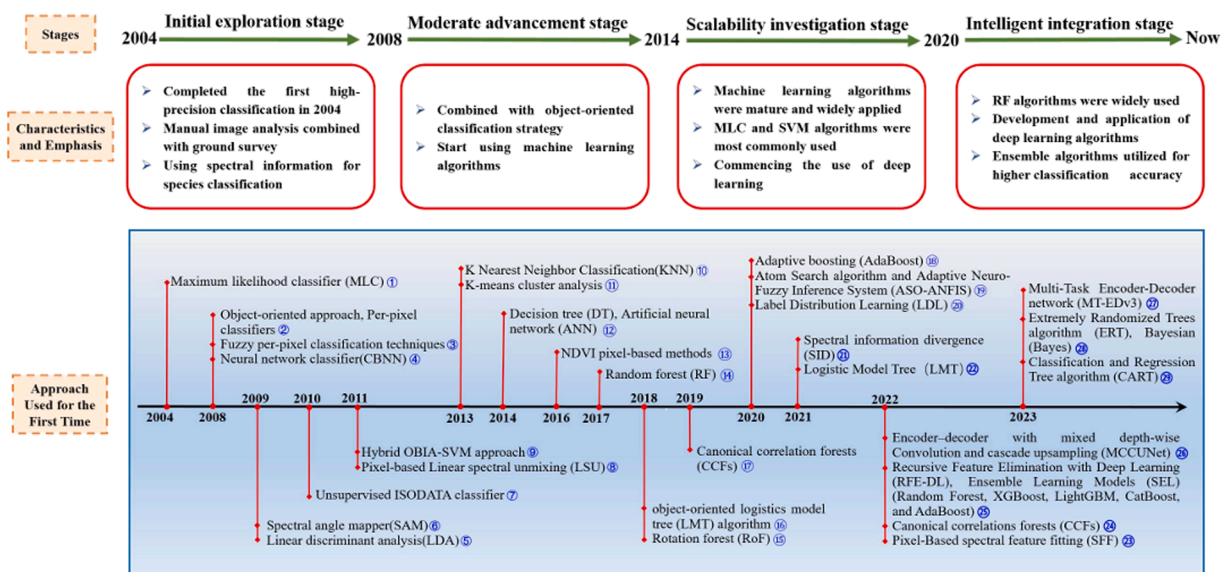


Fig. 5. Characteristics, emphasis and approach used for the first time in mangrove species mapping for different stage (the approach used for the first time in studies are referenced by ①(Wang et al., 2004b); ②(Myint et al., 2008); ③(Neukermans et al., 2008); ④(Wang et al., 2008); ⑤(Wang and Sousa, 2009); ⑥(Yang et al., 2009); ⑦(Kovacs et al., 2010); ⑧(Kamal and Phinn, 2011); ⑨(Heumann, 2011); ⑩(Zhang et al., 2013); ⑪(Manjunath et al., 2013); ⑫(Wong and Fung, 2014); ⑬(Candra et al., 2016); ⑭(Rogers et al., 2017); ⑮(Zhang et al., 2018); ⑯(Pham et al., 2018); ⑰(Pham et al., 2019a); ⑱(Peng et al., 2020); ⑲(Pham et al., 2020); ⑳(Xia et al., 2020); ㉑(Zulfa et al., 2021); ㉒(Cao et al., 2021); ㉓(Rahmandhana et al., 2022); ㉔(Wang et al., 2022a); ㉕(Fu et al., 2022); ㉖(Li et al., 2022); ㉗(Lassalle et al., 2023); ㉘(Wang et al., 2023); ㉙(Tian et al., 2023)).

3.4. Classification approaches of mangrove species mapping

Historically, the development of mangrove species mapping approaches based on remote sensing data was closely related to the development of statistical learning. Relevant literature published up to 2023, as summarized in Fig. 5, outlines all the approach used for the first time in mangrove species mapping. In the context of mangrove mapping, image classification methods can be broadly categorized into four distinct stages. In the initial exploration stage, manual image interpretation and spectral information extraction were utilized for species classification, with ground surveys are conducted for validation. Maximum likelihood classification (MLC) was the first approach used for identifying mangrove species distribution. As one of the most widely used supervised algorithms, MLC achieved classification overall accuracies ranged from 80 % to 98.6 % (Giri et al., 2014; Hirata et al., 2014).

With the advancement of image segmentation technology, object-oriented classification was first applied to mangrove species mapping in 2008, signifying the transition of this field into a moderate advancement stage. Unlike pixel-based classification, which classifies each pixel based solely on spectral information, object-oriented classification offers not only additional spectral details, such as mean, minimum, maximum, median, and variance values per band, but also spatial and texture information for objects (Blaschke, 2010). The OA acquired by object-oriented classification has demonstrated superior performance in mangrove species mapping compared to traditional pixel-based approaches (Dronova, 2015). For example, Myint et al. (2008) effectively identified three mangrove species using object-oriented classification, achieving a producer's accuracy of over 91 %, in contrast to the 51 % accuracy obtained with pixel-based classification. Similarly, Kamal and Phinn (2011) assessed both pixel-based and pixel-based approaches in mangrove species mapping, demonstrating the superiority of the object-oriented method. Despite these advancements, pixel-based supervised and unsupervised classification methods continue to be extensively employed for mapping mangrove species during this stage. Supervised classification methods like the spectral angle mapper (SAM) and MLC, necessitate either training data or prior knowledge of the study area. MLC and SAM classifier were successfully applied to multispectral images before 2009 and have been widely utilized in mapping mangrove species (Wang et al., 2004b; Kamal and Phinn, 2011). In comparison, unsupervised classification methods do not require training data or prior knowledge, which primarily include vegetation index (NDVI) based classification, K-means clustering algorithm, and iterative selforganizing data analysis techniques algorithm (ISODATA). During this phase, traditional statistical analysis approaches such as clustering-based neural network classifier (CBNN), linear discriminant analysis (LDA) and linear spectral unmixing (LSU), were progressively supplanted by machine learning algorithms.

During the scalability investigation stage, machine learning algorithm were widely adopted and matured in their usage. The combination of object-oriented machine learning approaches such as support vector machine (SVM) and random forest (RF), has demonstrated greater suitability for high spatial resolution data (Kumar et al., 2013; Wang et al., 2018b). These methods have the advantages of not requiring assumptions about data distribution and accommodate the simultaneous input of multiple variables (spectral, spatial, texture features and spatial information, etc.) to classify the target (Maurya et al., 2021). The majority of remote sensing research utilizing machine learning techniques to map three or more mangrove species has shown an accuracy spectrum ranging from 69.44 % to 97.24 %. Over 33 % of studies devised up to 2023 have utilized either SVM or RF for mangrove species mapping, achieving an average OA of 84 %. Nevertheless, there are certain limitations of traditional machine learning algorithms in mangrove species research (Chen et al., 2023). On the one hand, most machine learning algorithms are limited to single-task performance, with classification features often restricted to shallow image features derived from statistical data or simple nonlinear correlations. On the other hand, in certain task scenarios, the fine-tuning of machine learning models heavily relies on the specific domain expertise and personal experience of the developers, which limits the transferability to a large extent.

Since 2020, the most recent phase of mangrove species mapping research has increasingly emphasized automating image processing, particularly through the adoption of deep learning and ensemble learning methods. Deep learning is considered the state-of-the-art in machine learning, as it has the capability to overcome the limitations inherent in traditional machine learning approaches. Ensemble learning methods, which combine multiple basic learners together and take full advantage of them, often result in improved performance (Kattenborn et al., 2021). For instance, a recent study comparing Convolutional Neural Network (CNN), RF, and SVM in the classification of five mangrove species found that CNN significantly outperformed RF and SVM with an average OA of 88 % (Li et al., 2021). Lassalle et al. (2023) employed multiple images and algorithms, including Multi-Task Encoder-Decoder (MT-EDv3) network, SAM, MLC, and RF, to classify three mangrove species, achieving the highest OA of 97 % with MT-EDv3 algorithm. Similarly, Li et al. (2022) implemented a novel deep-learning algorithm algorithms (MCCUNet) on high-resolution UAV multispectral images to differentiate three mangrove species in Qinzhou City of China, achieving a peak OA of 97.24 %. In addition, Fu et al. (2022) evaluated the ability of deep learning models and stacking ensemble learning (SEL) algorithms in classifying mangrove species, indicating that the deep learning algorithm achieved an OA of 94.8 % and SEL algorithm outperformed the base models by 0.8–4.2 %.

4. Discussion

Despite significant progress has been made in mapping mangrove species, numerous aspects still require further investigation for enhancement. Given the above, the challenges and future pathways concerning data sources, classification algorithms, increased number of classification species, large-scale and temporal series mangrove species mapping has been discussed.

4.1. Multiple sensors for data acquisition

Mangrove species are often mixed with other species that have similar spectral reflectance (Md Isa and Suratman, 2021). With the growing number and spatial heterogeneity of mangrove species, subtle deviations between different species cannot achieve feasible or

desirable classification results using only limited spectral information from multispectral images (Zhang et al., 2014; Flores-de-Santiago et al., 2016). Therefore, integrating different types of data is crucial. The impetus in space exploration, coupled with the advancement made in technology, has brought us a plethora of remote sensing data, increasing opportunities for multi-sensor and multi-resolution data integration (Ghamisi et al., 2019; Kulkarni and Rege, 2020).

Effectively integrating remote sensing data that provide support in terms of spectral, geometric, texture, three-dimensional structural features, and features derived from other data sources can enrich information and improve classification accuracy. For example, in highly diversified forests, species may intermingle both across horizontal and vertical layers. A promising approach to improve this situation is to consider the three-dimensional structural features of LiDAR data, as it can effectively distinguish different species with significant structural differences (Li et al., 2021; Mäyrä et al., 2021; Zhang and Liu, 2013). Moreover, launching satellites with suitable spatial and spectral resolution, along with frequent revisits, enables the acquisition of multi-temporal information, such as phenological features (Dronova and Taddeo, 2022). For some mangrove species, their crowns may change significantly with the seasons, such as during flowering. This advancement also shows great potential in enhancing mapping results (Ayushi et al., 2022; Ni et al., 2021).

Although multi-source collaboration has become a common trend, there are limitations in fusing heterogeneous data (Swain et al., 2023). Additionally, data preprocessing poses significant challenges to researchers. Due to different imaging principles, structural differences could result in spatial and spectral distortion during fusion (Ghassemian, 2016; Toutin, 2004). Likewise, speckle noise and registration errors are inevitable. Thus, accurate and efficient data fusion remains a challenge. Moreover, current studies have not yet produced matching algorithms that achieve the highest accuracy for mangrove species mapping across different data sources. To address the inconsistencies in multi-source data, it is essential to develop automatic registration and mosaicking techniques tailored to heterogeneous datasets. Precise alignment of datasets with varying resolutions can be achieved using feature extraction and matching algorithms, eliminating edge discrepancies and enabling seamless mosaicking. Simultaneously, strengthening preprocessing steps, such as radiometric and atmospheric corrections, is crucial to enhance the radiometric consistency across different sensors. Building on these improvements, developing robust fusion algorithms, such as deep learning-based multi-modal data fusion frameworks, may maximize the use of diverse data characteristics while mitigating the effects of heterogeneity.

High-resolution remote sensing datasets (such as hyperspectral and LiDAR data) contain vast amounts of information, require substantial storage space, and incur high acquisition costs when applied to large areas. The introduction of UAV remote sensing platforms in recent years has opened up new avenues, as they are cost-effective and flexible solution for small-scale data acquisition (Adão et al., 2017). The future holds significant potential for integrating various high spatial and spectral resolution optical imageries, SAR and LiDAR, along with multi-sensor, multiresolution, and temporal data (Fassnacht et al., 2016; Wan et al., 2021).

4.2. Classification and accuracy improving strategy

Research on mangrove species mapping based on remote sensing have applied a diverse array of classification algorithms (see Section 3.4). Although the early pixel-based classification process is simple and straightforward, it is prone to the salt-and-pepper phenomenon. Conversely, object-oriented classification methods, considering the spatial structure and contextual information of the image, can offer improvements in both accuracy and stability of the classification results. Therefore, image segmentation gradually shifts from pixel-based to object-oriented classification methods (Kucharczyk et al., 2020). With the continuous advancement of digital imaging technology, ensemble learning and deep learning methods have been shown to significantly enhance classification accuracy through more complex feature extraction and model combination strategies (Fu et al., 2022; Li et al., 2022). With the improvement of image resolution and the continuous optimization of classification algorithm, the process of species mapping is progressively shifting towards automation (Yuan et al., 2021). However, several challenges have arisen in classification strategies that need further improvement.

One major challenge lies in feature extraction and selection (Farmonov et al., 2023). For example, hyperspectral data provides abundant spectral information for species classification, but data processing poses certain difficulties (Sun and Du, 2019). Previous researches has demonstrated that a substantial volume of data features can result in the curse of dimensionality, leading to feature redundancy and a reduction in classification accuracy (Keogh and Mueen, 2017; Jia et al., 2022). Given this challenge, it becomes essential to devise advanced deep feature learning models aimed at uncovering the complex patterns and distinctive information embedded within high-dimensional datasets. Additionally, model selection and training also stand as a crucial challenge in this process as the significant constraints posed by their high computational demands and low training efficiency.

Taking advantage of digital image processing technology and deep learning algorithms, classification can become more convenient and accurate (Zhong et al., 2018). However, owing to the disparity between the irregular objects derived from image segmentation and the rectangular classification units necessitated by deep learning, the combination of object-oriented and deep learning for mangrove species classification still needs further exploration (Kumari and Kaul, 2023; Kucharczyk et al., 2020). Moreover, the most notable limitation of these algorithms is their limited transferability, and the scarcity and lack of representativeness of sample points can also reduce the generalizability of classification models, especially when applied across different ecological zones. Overcoming these challenges remains a priority for researchers to improve the effectiveness and adaptability of classification algorithms in mangrove species mapping.

When the number of mangrove species to be differentiated exceeds seven, accurately distinguishing different species and achieving high classification accuracy becomes a challenge. In such cases, leveraging phenological characteristics to highlight species differences, exploring new methods and techniques to effectively extract and identify feature differences between images, or increase the number of high-quality training and validation samples would be feasible approaches to significantly improve classification accuracy.

High-quality training and validation samples are fundamental to improving classification accuracy. However, due to the remote and complex terrain of mangrove regions, sample collection is often constrained by technical and logistical limitations. To address this challenge, meta-analysis methods could provide a valid way to integrate sample data from existing studies, thus alleviate the limitations posed by insufficient sample points. Furthermore, establishing an open and shared mangrove species sample database would provide researchers with reliable access to high-quality sample data, fostering advancements in this field.

4.3. Large-scale and long-term series mapping

It is noteworthy that the majority of research on remote sensing-based mangrove species classification has primarily focused on analyzing small-area regions relying on mono-temporal remote sensing data and field observations. Analysis of all literature reveals that approximately 92 % of the research utilizes single-temporal remote sensing imagery, while studies employing multi-temporal or long-term time series data account for only about 8 %. However, conducting mangrove species research on a broader scale and long time series proves to be of paramount significance for mangrove management and policy development. Large-scale species mapping can provide detailed and consistent information to aid in the conservation and sustainable management of these ecosystems. Globally, only a small fraction of mangrove species has been mapped using multiple remote sensing datasets (Fig. 4). Location accessibility, local and regional governments approval, and resource availability (e.g., instruments, time, and funding) likely explain the distribution and frequency of study sites. More often, the same research areas are revisited using different remote sensing sensors. While revisiting is essential for monitoring change in mangroves species, the results of the reviewed articles indicated, however, that monitoring species change and temporal series mapping has been of lower interest in many mangrove habitats. The key to long-term monitoring lies in analyzing longitudinal data to detect changes in mangrove species composition and health status over time. The limited utilization of multi-temporal and long-term time-series datasets can be addressed by integrating public long-term datasets. Additionally, by exploring advanced time-series analysis techniques, such as machine learning-based time-series models, it becomes possible to extract meaningful temporal patterns from multi-temporal datasets and identify subtle temporal variations associated with species, thereby improving the accuracy of species time-series analysis.

To date, mangrove species mapping have expanded from individual conservation areas to regional scales, such as the Hainan Island of China (Xue et al., 2024). However, there is still a lack of mangrove species mapping at continental or global scales. Mangroves grow in tropical regions, and while advanced technologies like LiDAR scanners and hyperspectral sensors can capture detailed species information, achieving large-scale species mapping with these technologies is challenging compared to the broad coverage capabilities of medium- to high-resolution satellite data. This challenge is primarily due to the time-consuming nature, high labor intensity, and significant cost of LiDAR and hyperspectral sensors (Chen and Sun, 2024). Studies indicates that medium- to high-resolution remote sensing data are effective for large-scale species identification. For instance, researchers have utilized Sentinel-2 (Fang et al., 2023; Xie et al., 2021) and integrated data from Sentinel-1/2 (Bhattacharai et al., 2021; Liu et al., 2023) or MODIS, Landsat and PALSAR-2 (Zhang et al., 2024) to extract species information on a large scale.

Satellite data offers advantages in terms of ease of acquisition and management and can be periodically updated at specific intervals, providing ecologists with a continuous and stable observation method. Thus, publicly accessible remote sensing imagery serves as a dependable and efficient tool for covering extensive areas and delivering long-term species data. Therefore, open remote sensing and data processing platforms, such as Google Earth Engine (GEE), are essential (Amani et al., 2020). As a comprehensive platform, GEE integrates cloud computing, image analysis, and geospatial data visualization, enabling finer temporal resolution studies by streamlining the process of using open remote sensing data sets such as the complete archive of MODIS, Landsat and Sentinel (Tamiminia et al., 2020; Velastegui-Montoya et al., 2023). Additionally, cloud cover and tidal disturbances can affect the images, but GEE can conveniently select optimal images. Cloud platforms like GEE are closely connected to large-scale and temporal series species mapping.

5. Conclusions

Considering the importance of developing mangrove species classification research for biodiversity conservation and sustainable environmental management, this study conducted a bibliometric and descriptive statistics visualization analysis on previous research. As one of the main research directions of mangrove researchers, the number of publications on mangrove species mapping has generally shown an upward trend. It performs a statistical analysis of the keywords in the literature and visualizes the important subfields identified through clustering. The results indicate that remote sensing technology and machine learning are central to the 12 subfields. The network of cooperation among countries shows that China, the United States, and India are the three countries with the most research on mangrove species mapping, with the United States scholars notably engaging in more international collaborations. The results provide scholars with valuable insights into which countries are active in initiating cooperative research and encourage scholars to actively seek international cooperation opportunities in this domain. Most studies focus on small areas such as nature reserves, with Asia (67.92 %) and North America (13.21 %) being the two continents with the most research areas, and the frequently studied sites primarily located in India (9 times) and China (8 times). In terms of remote sensing sensors, satellite, aircraft and UAV remote sensing data are predominantly used. The rapid development of UAV remote sensing platforms, with their flexibility, high resolution, and low cost, provides more opportunities for mangrove species mapping. For image processing and classification, early studies primarily used object-oriented segmentation methods combined with machine learning algorithms. In the latest stage, with the continuous advancements in artificial intelligence technologies, classification algorithms based on ensemble learning or deep learning have shown significant improvement in classification accuracy than machine learning, and are expected to become important research

directions in the future. Undoubtedly, future research in mangrove species mapping based on remote sensing is advancing towards refining information extraction, automation, and intelligence. This involves integrating multiple data sources, fusing multi-temporal data, and incorporating various classification algorithms. Future research should focus on developing robust multi-sensor data fusion methods, such as deep learning-based multi-modal frameworks, to maximize the complementary strengths of diverse datasets while addressing heterogeneity challenges. Additionally, improving temporal series analysis by leveraging platforms like GEE can streamline large-scale mapping and enhance the monitoring of species changes over time with higher temporal resolution. This article serves as a reference for researchers and practitioners from various fields attempting to apply remote sensing data to mangrove species mapping and monitoring, inspires future applications of remote sensing technology in this field.

Ethical statement

Not applicable: This manuscript does not include human or animal research. If this manuscript involves research on animals or humans, it is imperative to disclose all approval details.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.gecco.2025.e03408](https://doi.org/10.1016/j.gecco.2025.e03408).

Data availability

Data will be made available on request.

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